

Extremality shift of rotating black holes at finite Gauss–Bonnet coupling

Abstract: We study how the minimum mass of a rotating black hole at fixed angular momentum changes with the Gauss–Bonnet coupling in five dimensions. We construct stationary, asymptotically flat solutions with two equal angular momenta at finite coupling and calculate the thermodynamic potential Ψ conjugate to that coupling. The calculation differentiates the discretised field equations implicitly and solves a linear response problem; it avoids real finite differences between neighbouring solutions, but remains numerical. Extrapolating to zero temperature gives the local slope $\partial M_{\text{ext}}/\partial\alpha|_J$, subject to a regular extremal limit. Across the sampled range this slope stays positive: it decreases from the known perturbative value π to about 1.07149, then rises to 1.44440. Thus the extremal mass increases monotonically while its slope does not. The potential and the changes in the extrapolated mass curve satisfy an integral consistency check, with shared numerical inputs and errors. We validate the calculation against the analytic static solution, the published perturbative potential and the extremal near-horizon entropy. We report sensitivity to the extrapolation model, fitting window and temperature coordinate, and restrict the interpretation to classical Einstein–Gauss–Bonnet theory in the range computed.

Keywords: Black Holes, Classical Theories of Gravity, Higher Curvature Gravity

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1 Introduction

For a rotating black hole, extremality marks the limiting mass at a given angular momentum on the branch under consideration. Changing the gravitational theory changes that limit. The

question addressed here is quantitative: how does the extremal mass respond when the coefficient of a Gauss–Bonnet term is increased, beyond an infinitesimal perturbation of Einstein gravity?

We study five-dimensional, asymptotically flat black holes with two equal angular momenta, $J_1 = J_2 = J$. The equality of the spins reduces the metric equations to ordinary differential equations in the radial coordinate. At zero coupling the solution is Myers–Perry [5]. For a Gauss–Bonnet term with the normalisation used below, the extremal mass is known to first order [8]:

$$M_{\text{ext}} = \frac{3}{2}\pi^{1/3}J^{2/3} + \pi\alpha + \mathcal{O}(\alpha^2). \quad (1)$$

Here J is the angular momentum in each plane. The initial slope is positive, but this expression does not determine the slope at finite coupling. The perturbative thermodynamics can be obtained without the corrected metric using the method of Reall and Santos [4]; relevant results for Myers–Perry black holes and their metric corrections are given in [6, 7].

The underlying finite-coupling solutions are also known numerically. Brihaye, Kleihaus, Kunz and Radu [9] constructed the equal-spin family and derived its extremal near-horizon entropy using Sen’s entropy function [10]. Extremal bulk solutions and their domain of existence were studied further in [11]. Our focus is the response of this family in the coupling direction, including the finite-coupling dependence of the extremal mass slope.

We use the extended first law

$$\mathrm{d}M = T \mathrm{d}S + 2\Omega_H \mathrm{d}J + \Psi \mathrm{d}\alpha. \quad (2)$$

The quantity Ψ is a thermodynamic response coefficient for a whole solution, not an additional radial field. It measures the change of mass per unit change of coupling at fixed entropy and angular momenta. Extended first laws for Lovelock couplings are developed in [12, 20]; the latter explicitly treats stationary black holes. Varying α compares equilibrium solutions of theories with different couplings. It does not describe a time-dependent change of the coupling in a single spacetime.

The computation has three steps. We first solve the nonlinear radial boundary-value problem at fixed horizon radius, angular velocity and coupling. We then differentiate that discretised problem to determine how its solution and charges respond to α , and use (2) to extract Ψ . Finally, we follow the family towards zero temperature and extrapolate Ψ . On a regular extremal branch the first law identifies that limit with $\partial M_{\text{ext}}/\partial\alpha|_J$. The thermodynamic identities are analytic; the rotating finite-coupling solutions, their responses and the extremal extrapolation are numerical.

The measured potential changes sign between slowly rotating and near-extremal states. Its extremal limit stays positive over the sampled couplings, falls substantially below π , and then rises again. The result concerns the slope of the mass, not a non-monotonic mass. We check this interpretation against changes in the extrapolated mass curve, while accounting for the numerical inputs shared by the two calculations.

Extremality shifts are often discussed in connection with the weak gravity conjecture [1, 2] and the thermodynamic entropy–extremality relation of Goon and Penco [3]. For the present rotating, uncharged family we use that relation as a thermodynamic identity, without identifying angular momentum with electric charge. Our finite-coupling result is a statement about classical Einstein–Gauss–Bonnet gravity. Its interpretation as a gravitational effective field theory is restricted, as discussed in section 6.

Section 2 specifies the solutions and observables. Section 3 explains the potential and its numerical evaluation. Results and checks follow in sections 4 and 5; appendix A describes the numerical construction and the sensitivity of the extrapolation.

2 The equal-spin family and its thermodynamics

2.1 Action and ansatz

We use

$$I = \frac{1}{16\pi G} \int d^5x \sqrt{-g} (R + \alpha \mathcal{L}_{\text{GB}}), \quad \mathcal{L}_{\text{GB}} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, \quad (3)$$

and set $G = 1$. The coupling has dimensions of length squared. We restrict the study to $\alpha \geq 0$ and the branch connected continuously to Einstein gravity. This sign choice is a restriction of the study, not a consequence of the graviton propagator about flat space: the Gauss–Bonnet contribution to the field equations is quadratic in curvature and vanishes at linear order about Minkowski space. The equations are $G_{\mu\nu} + \alpha H_{\mu\nu} = 0$, with $H_{\mu\nu}$ the Lanczos tensor, and contain at most second derivatives [14].

The equal-spin ansatz [9] is

$$ds^2 = -b(r)dt^2 + \frac{dr^2}{f(r)} + r^2[d\theta^2 + \sin^2\theta \cos^2\theta (d\varphi_1 - d\varphi_2)^2] + h(r)[\sin^2\theta d\varphi_1 + \cos^2\theta d\varphi_2 - w(r)dt]^2. \quad (4)$$

Here $\theta \in [0, \pi/2]$ and $\varphi_i \in [0, 2\pi)$. The four radial functions b, f, h, w are unknown. The horizon is at $r = r_H$, where $b = f = 0$, and its angular velocity is $\Omega_H = w(r_H)$. Asymptotic flatness requires $b, f, h/r^2 \rightarrow 1$ and $w \rightarrow 0$. Regularity and these boundary conditions define the numerical problem. The solutions are stationary: their metric is independent of time, although their horizons rotate.

We set $r_H = 1$ in the numerical construction and label the rotation by

$$q = r_H \Omega_H. \quad (5)$$

At $\alpha = 0$, q ranges from zero to $1/\sqrt{2}$ at extremality. At finite coupling the extremal value changes. Statements about the physical coupling can instead be made in the scale-invariant variables below; the tabulated numerical α is α/r_H^2 in these units.

2.2 Charges, entropy and the first law

The asymptotic coefficients determine the mass and each angular momentum:

$$b = 1 + \frac{U}{r^2} + \mathcal{O}(r^{-4}), \quad w = \frac{W}{r^4} + \mathcal{O}(r^{-6}), \quad M = -\frac{3\pi}{8}U, \quad J = \frac{\pi}{4}W. \quad (6)$$

The normalisations agree with [9]. The horizon temperature is $T = \sqrt{b'(r_H)f'(r_H)}/(4\pi)$. We use Wald entropy [15, 16], which for this stationary Lovelock horizon has the Jacobson–Myers form [17]:

$$S = \frac{1}{4} \int_{\mathcal{H}} d^3x \sqrt{\sigma} (1 + 2\alpha \tilde{R}), \quad \tilde{R} = \frac{8}{r_H^2} - \frac{2h(r_H)}{r_H^4}. \quad (7)$$

Thus S depends on α both explicitly and through the geometry. The intrinsic expression applies to the stationary horizon cross-section; we do not use it as an entropy prescription for dynamical horizons.

The two equal angular momenta contribute $\Omega_1 dJ_1 + \Omega_2 dJ_2 = 2\Omega_H dJ$ to (2). Scaling the lengths yields

$$2M = 3TS + 6\Omega_H J + 2\alpha\Psi. \quad (8)$$

We evaluate this relation as an internal consistency check. For $\alpha \neq 0$ it also gives $\Psi = (2M - 3TS - 6\Omega_H J)/(2\alpha)$, but subtraction in its numerator is ill-conditioned as $\alpha \rightarrow 0$. At $\alpha = 0$ it cannot determine Ψ . The response calculation below avoids this division.

2.3 Scale-invariant variables

Under $r_H \rightarrow \lambda r_H$, the quantities (M, α) scale as λ^2 , (S, J) as λ^3 and (T, Ω_H) as λ^{-1} . The potential Ψ is dimensionless. For nonzero spin we define

$$y = \frac{\alpha}{J^{2/3}}, \quad \tau = T J^{1/3}, \quad \mu = \frac{M}{J^{2/3}}, \quad \sigma = \frac{S}{J}. \quad (9)$$

On the extremal branch, scale invariance implies $M_{\text{ext}}(J, \alpha) = J^{2/3} \mu(y)$, hence $\partial M_{\text{ext}} / \partial \alpha|_J = \mu'(y)$. This rescaling lets solutions computed at fixed r_H describe the extremal mass at fixed J .

Since τ vanishes both at extremality and as $J \rightarrow 0$ in the static limit, it is unsuitable as a single coordinate for the whole family. For the non-extremal plots we use [11]

$$x = \frac{3\pi\alpha}{4M}, \quad j = \left(\frac{3}{2}\right)^{3/2} \sqrt{\pi} \frac{J}{M^{3/2}}, \quad t_H = 4\sqrt{\frac{2\pi}{3}} T \sqrt{M}. \quad (10)$$

The normalisation gives $t_H = 1$ for the static Einstein solution and $j = 1$ for extremal Myers–Perry. The measured Einstein extremal value is $j = 1.00000000$. At finite coupling the static temperature is instead $t_H = \sqrt{1 + 2\bar{\alpha}} / (1 + 4\bar{\alpha})$, where $\bar{\alpha} = \alpha / r_H^2$; it is not unity.

3 What the Gauss–Bonnet potential measures

3.1 Definition and evaluation along the numerical family

By definition, Ψ is the derivative of mass with respect to α at fixed entropy and both angular momenta. The solver, however, fixes r_H and Ω_H . Along its coupling direction, both entropy and angular momentum generally change. Evaluating the first law on that direction gives

$$\boxed{\Psi = \left. \frac{\partial M}{\partial \alpha} \right|_{S, J} = \left. \frac{\partial M}{\partial \alpha} \right|_{r_H, \Omega_H} - T \left. \frac{\partial S}{\partial \alpha} \right|_{r_H, \Omega_H} - 2\Omega_H \left. \frac{\partial J}{\partial \alpha} \right|_{r_H, \Omega_H}}. \quad (11)$$

The subtraction removes the mass changes associated with entropy and spin, leaving the response to the coupling itself. Equation (11) is therefore valid for the rotating stationary solutions, not only for the static member. It does not equate partial derivatives with different constraints: it equates one constrained derivative to a combination of three derivatives evaluated along a different path.

No terms such as $S \partial_\alpha T$ appear, because we evaluate the differential first law; we do not differentiate a putative expression $M = TS + 2\Omega_H J$. More generally any differentiable path through equilibrium solutions with $d\alpha \neq 0$ gives the same Ψ after subtracting the entropy and spin contributions and dividing by $d\alpha$.

3.2 Connection with extremality

Suppose the sampled family approaches a regular extremal branch with $S = S_{\text{ext}}(J, \alpha)$. Differentiating $M_{\text{ext}}(J, \alpha) = M(S_{\text{ext}}(J, \alpha), J, \alpha)$ gives

$$\left. \frac{\partial M_{\text{ext}}}{\partial \alpha} \right|_J = T \left. \frac{\partial S_{\text{ext}}}{\partial \alpha} \right|_J + \Psi = \Psi_{\text{ext}}, \quad T = 0. \quad (12)$$

To infer this value from non-extremal numerical solutions we assume that their potentials have a continuous limit on that branch and that the entropy term has a vanishing limit. A bounded extremal entropy derivative is a sufficient regularity condition. Agreement with the algebraic near-horizon entropy and smooth sampled data supports this assumption over the range studied, but a finite table is not a proof of it.

At positive temperature the same first law implies the exact identity

$$\left. \frac{\partial S}{\partial \alpha} \right|_{M,J} = -\frac{\Psi}{T}.$$

Its extremal limit is the entropy–extremality relation of Goon and Penco [3]. In particular, positive Ψ means decreasing entropy at fixed mass and angular momenta. This should not be confused with the entropy along the extremal branch, where the mass also changes with α .

3.3 Numerical response by implicit differentiation

Let u collect the metric amplitudes at the N collocation nodes. The discretised equations and boundary conditions are $R_N(u; \alpha, r_H, \Omega_H) = 0$. After solving this nonlinear system, we differentiate it at fixed r_H, Ω_H :

$$\underbrace{\frac{\partial R_N}{\partial u}}_{\mathbf{A}} \frac{\partial u}{\partial \alpha} = -\frac{\partial R_N}{\partial \alpha}. \quad (13)$$

Where the Jacobian $\mathbf{A}$ is invertible, one linear solve gives the response of the discretised metric. Propagating it through the same charge and entropy functionals gives the three derivatives in (11). The entropy response retains the explicit coupling derivative of (7), in addition to the change of the horizon geometry.

The production code assembles $\mathbf{A}$ by the chain rule from local complex-step derivatives of the radial equations and horizon relations. The parameter derivative on the right is also evaluated by complex step. For a real analytic function, $\text{Im } f(p + i\epsilon)/\epsilon = f'(p) + \mathcal{O}(\epsilon^2)$. A small imaginary step avoids the subtraction of nearly equal real numbers. The implementation and its verification are described in appendix A.

Equation (13) is an analytic differentiation identity for the discrete system. Its numerical solution still carries discretisation, nonlinear-solver, linear-solver and rounding errors. The extracted observables also carry asymptotic fitting errors. We consequently refer to this procedure as implicit numerical response, rather than an exact finite-coupling determination of Ψ .

This local linear response is evaluated about solutions at finite α . It does not truncate the whole family at first order about $\alpha = 0$. That distinction is what allows us to measure a non-perturbative dependence on the coupling within classical EGB gravity.

As a check, we compare the response with central real differences of neighbouring solutions. At 4 points, using 5 step sizes related by halving, the discrepancy decreases by factors between 3.951 and 4.028, consistent with second-order convergence. The largest initial gap is 1.47×10^{-2} . Responses in both α and Ω_H also provide tangent predictors for the continuation towards extremality.

4 Results

The dataset contains 356 accepted solutions at 13 couplings from $\alpha = 0$ to $\alpha = 1/2$ in units $r_H = 1$. It reaches $x = 0.5001$ on the sampled family and $x = 0.4170$ on the extrapolated extremal branch. These are subsets of the allowed domain, whose published bound is $x < 1$ [11]. We follow the branch continuously connected to Myers–Perry. Continuation and regularity checks identify the branch sampled here; they do not establish global uniqueness or exhaust its domain.

4.1 An analytic example: the static limit

The static member is the Boulware–Deser solution [13]. In our conventions,

$$M = \frac{3\pi}{8}(r_H^2 + 2\alpha), \quad T = \frac{r_H}{2\pi(r_H^2 + 4\alpha)}, \quad S = \frac{\pi^2}{2}r_H^3 + 6\pi^2\alpha r_H. \quad (14)$$

This example also illustrates the constraints in (11). At fixed radius $M_\alpha = 3\pi/4$ is positive, but the entropy changes by $S_\alpha = 6\pi^2 r_H$. Subtracting its contribution gives

$$\Psi_{\text{static}} = \frac{3\pi}{4} - T(6\pi^2 r_H) = \frac{3\pi}{4} \frac{4\bar{\alpha} - 3}{1 + 4\bar{\alpha}}, \quad \bar{\alpha} = \frac{\alpha}{r_H^2}. \quad (15)$$

In particular $\Psi_{\text{static}} \rightarrow -9\pi/4$ as $\alpha \rightarrow 0$. The mass derivative at fixed radius and the derivative at fixed entropy can have opposite signs; keeping the entropy fixed requires the radius to change. Equation (15) satisfies (8) identically at zero spin.

Applying the numerical procedure at zero spin for 7 couplings reproduces the analytic mass, temperature, entropy and potential to worst relative deviations 3.20×10^{-11} , 1.08×10^{-13} , 1.00×10^{-16} and 2.60×10^{-9} , respectively. These are checks against an analytic solution at finite coupling. The potential changes sign at $\bar{\alpha} = 3/4$, or $x = 3/5$, on the static branch. That result alone does not fix the sign on all rotating solutions beyond the range sampled here.

4.2 The published perturbative potential

The full spin dependence at $\alpha = 0$ follows from the perturbative free energy in [6]. Define the Gibbs potential $\mathcal{G} = M - TS - \sum_i \Omega_i J_i$. The extended first law gives $d\mathcal{G} = -S dT - \sum_i J_i d\Omega_i + \Psi d\alpha$, so the coefficient of α in $\mathcal{G}$ at fixed T, Ω_i is Ψ_0 .

Specialising eq. (33) of the arXiv version of [6] to $D = 5$ and equal rotation parameters gives

$$\frac{\Delta\mathcal{G}}{\alpha} = -\frac{\pi}{4r_+^4} (a^4 - 14a^2 r_+^2 + 9r_+^4).$$

The Ricci-flat background makes the curvature-squared expression applicable to the Gauss-Bonnet term at this order. In our radial gauge $r_H^2 = r_+^2 + a^2$ and $q = a/r_H$, hence

$$\Psi_0(q) = -\frac{\pi}{4} (u^2 - 14u + 9), \quad u = \frac{a^2}{r_+^2} = \frac{q^2}{1 - q^2}. \quad (16)$$

This published analytic result provides a reference for the numerical calculation. It gives $-9\pi/4$ at $u = 0$ and π at the extremal value $u = 1$, and vanishes at $q = 0.634936$.

We evaluate the numerical response on the Myers–Perry background at $\alpha = 0$. It agrees with (16) to at least 8 significant digits at 9 of 11 sampled spins, with worst relative deviation 2.86×10^{-8} . No extrapolation in the coupling is needed, although the numerical response and charge extraction remain discretised.

4.3 The potential across the family

Figure 1 shows Ψ against t_H for the couplings sampled across the spin range. On those sequences Ψ is negative near the static member and becomes positive towards extremality. Thus increasing α lowers the mass at fixed (S, J) in the former region and raises it in the latter. The location of the sign change depends on the coupling.

Every crossing in table 1 is bracketed by two accepted solutions of opposite sign. The quoted position is an interpolant; the bracket width indicates the resolution of the crossing. At zero coupling its difference from the analytic zero is 1.95×10^{-4} . In the scaled spin the locus moves from $j = 0.98074$ to 0.47284 . In temperature it moves from $t_H = 0.32448$ down to 0.19519 at $\alpha = 0.1$, then increases. A turning point in this coordinate does not by itself establish a common physical mechanism with the minimum of the extremal slope.

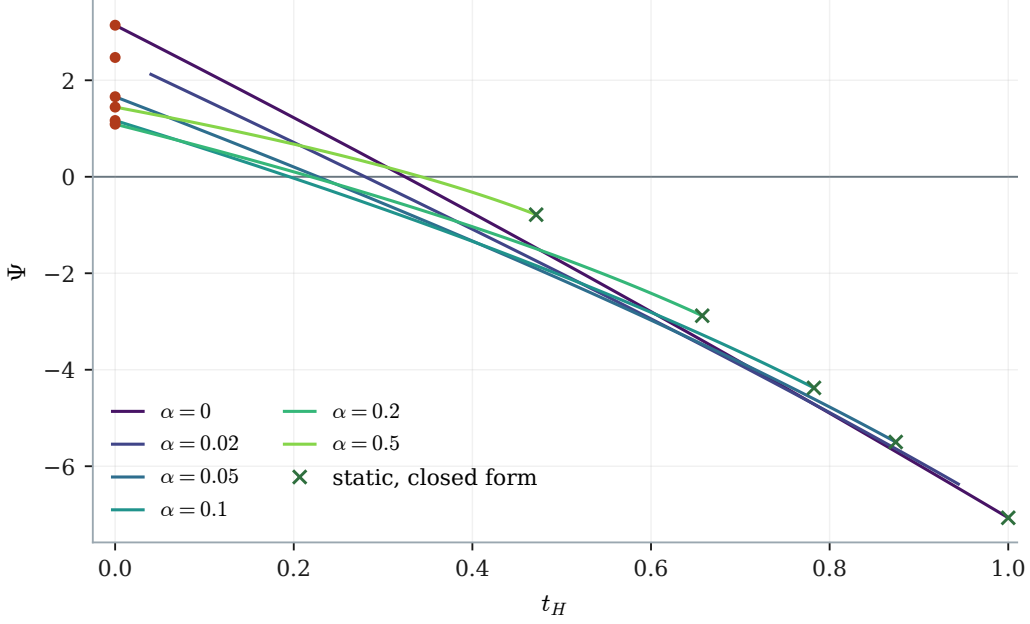


Figure 1. The potential along the sampled stationary families. The static endpoint depends on coupling: $t_H = \sqrt{1 + 2\bar{\alpha}}/(1 + 4\bar{\alpha})$. Crosses mark the analytic static values; circles at $t_H = 0$ mark extrapolated extremal values for the same couplings. Only the Einstein static endpoint has $t_H = 1$.

α	x	t_H	bracket width in t_H	j	q
0	0.0000	0.32448	7.35×10^{-2}	0.98074	0.63474
0.02	0.0247	0.27982	5.73×10^{-2}	0.94334	0.66943
0.05	0.0629	0.22716	4.93×10^{-2}	0.90205	0.71070
0.1	0.1240	0.19519	3.92×10^{-2}	0.85537	0.73172
0.2	0.2331	0.21922	2.82×10^{-2}	0.77828	0.67959
0.5	0.4829	0.34262	1.33×10^{-2}	0.47284	0.37294

Table 1. The measured sign-change locus. Digits of the interpolant are supplied for reproducibility, not as an error estimate. At $\alpha = 0$ the crossing is $q = 0.63474$, compared with the analytic $q = 0.634936$.

4.4 The extremal mass and its slope

The walks reach $\tau \leq 4.61 \times 10^{-3}$ at every coupling and 3.34×10^{-6} at the closest approach. The extrapolated branch is given in table 2. Central values are cubic fits in the unscaled temperature T to the six coldest states at each coupling. The sensitivity column includes changes of polynomial degree, fitting window, non-analytic fit terms and replacement of T by τ . The precise procedure is in appendix A. These are extrapolations from positive temperature, not solutions constructed directly at $T = 0$.

The main qualitative result is visible in figure 2: the slope Ψ_{ext} is positive throughout the sampled range, falls below its perturbative value, and subsequently increases. The lowest sampled central value is 1.07149 at $y = 0.1752$, with fit sensitivity 1.30×10^{-2} . Its neighbouring samples lie at $y = 0.118$ and 0.229. The last sampled value is 1.44440 at $y = 0.5112$. The suppression at the lowest sampled point relative to π is about a factor 2.93. We do not assign a precise location or confidence interval to a continuous minimum between the sampled couplings.

The mass itself remains increasing at fixed spin. The accumulated mass change is the integral

α	y	x	μ	Ψ_{ext}	sensitivity	j	σ
0	0.00000	0.00000	2.196888	3.14159	1.15×10^{-2}	1.00000	6.2832
0.005	0.00479	0.00510	2.211579	2.99094	1.12×10^{-1}	0.99005	6.6686
0.01	0.00984	0.01042	2.226279	2.82456	8.81×10^{-2}	0.98026	7.0685
0.02	0.02072	0.02165	2.255058	2.47283	1.60×10^{-1}	0.96156	7.8933
0.035	0.03843	0.03946	2.294438	2.00066	1.32×10^{-1}	0.93691	9.1213
0.05	0.05689	0.05758	2.328017	1.65908	1.86×10^{-2}	0.91671	10.2781
0.075	0.08775	0.08711	2.373482	1.32597	2.20×10^{-2}	0.89050	12.0267
0.1	0.11787	0.11521	2.410729	1.16621	2.10×10^{-2}	0.86994	13.5922
0.15	0.17520	0.16686	2.473993	1.07149	1.30×10^{-2}	0.83679	16.3436
0.2	0.22906	0.21316	2.531936	1.09007	1.12×10^{-2}	0.80823	18.7577
0.3	0.32921	0.29309	2.646582	1.20781	6.16×10^{-3}	0.75628	22.9735
0.4	0.42251	0.36002	2.765180	1.33384	6.96×10^{-3}	0.70815	26.6807
0.5	0.51124	0.41702	2.888535	1.44440	1.39×10^{-3}	0.66328	30.0608

Table 2. Extrapolated extremal states. The sensitivity refers to Ψ_{ext} , not to the other columns; it is an envelope of the fit variations described in the text, not a statistical confidence interval or a rigorous error bound. Extra digits identify the numerical central values. The first row is compared with $\mu = (3/2)\pi^{1/3}$ and $\Psi_{\text{ext}} = \pi$ in section 5.

of the slope:

$$M_{\text{ext}}(J, \alpha) - M_{\text{ext}}(J, 0) = J^{2/3} \int_0^{\alpha/J^{2/3}} \Psi_{\text{ext}}(y') dy'.$$

Thus the suppression of the local slope is not a suppression factor for the total mass shift.

We compare the potential with a second quantity formed from the extrapolated masses and spins. Scale invariance implies

$$\int_{y_1}^{y_2} \Psi_{\text{ext}}(y) dy = \mu(y_2) - \mu(y_1). \quad (17)$$

The right side does not use the entropy formula, so this provides a useful consistency test. It is not an independent determination: both sides share masses, spins, the solver and the temperature-based selection and extrapolation of the near-extremal states.

We integrate a local cubic interpolant through the four nearest potential samples. The largest interval discrepancy in (17) is 1.64×10^{-4} , or 1.96×10^{-3} relative to the corresponding change in μ . Across the full range it accumulates to 4.38×10^{-4} on a total change 0.6916. Using trapezoids instead gives a largest discrepancy 8.76×10^{-4} . The comparison includes quadrature error and correlated errors in the observables. Agreement of the changes in μ does not bound a common offset in the masses, and we do not use it as an absolute mass-accuracy estimate or a calibration of the sensitivity bars.

A one-sided quadratic fit to the lowest-coupling mass samples gives $\mu'(0) = 3.144521$, differing from π by 2.93×10^{-3} . Varying the number of samples gives a spread 2.09×10^{-2} . This derivative uses the mass curve rather than the entropy formula, but still relies on the thermodynamic extraction and extremal extrapolation.

5 Checks against known results

The analytic static and perturbative checks were described above. Here we compare the extremal mass and entropy with published results and distinguish those comparisons from internal consistency tests.

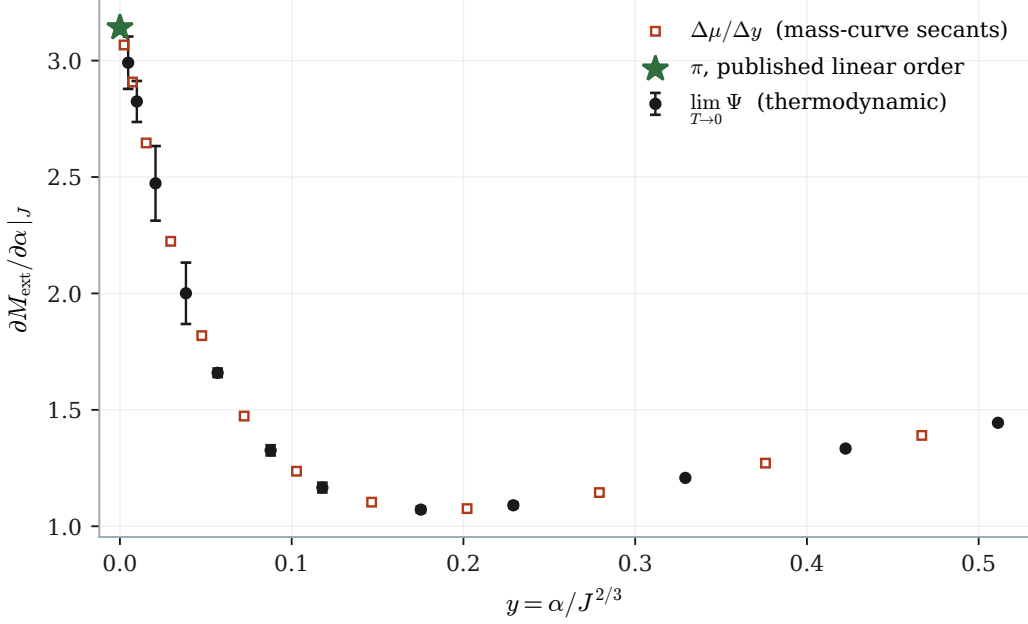


Figure 2. The extremal mass slope. Circles show the extrapolated potential and the fit-sensitivity envelopes; open squares show secants of $\mu(y)$ at interval midpoints. A secant is an interval average, not a pointwise derivative. These estimates share numerical inputs. The star is the published perturbative value π [8], which is not imposed in the response calculation.

Einstein and perturbative extremal limits. At $y = 0$ we obtain

$$\mu = 2.196887831, \quad \frac{3}{2}\pi^{1/3} = 2.196887831, \quad \Psi_{\text{ext}} = 3.141592534.$$

The absolute deviations from their respective analytic values are 4.56×10^{-11} and 1.19×10^{-7} , respectively. These numbers result from the same continuation and extrapolation used at finite coupling; the limiting constants are not fitted constraints. The agreement at $y = 0$ does not by itself establish the error at finite y .

Extremal entropy. The near-horizon entropy function of [9] determines S_{ext}/J as a function of y without integrating the radial equations. Our rederivation is summarised in appendix B. Solving its algebraic equations numerically at finite coupling gives the comparison in figure 3; the worst relative difference from the extrapolated bulk entropy is 4.06×10^{-6} . This validates the extremal entropy at finite coupling, but does not provide a near-horizon measurement of the asymptotic mass.

In the variables of [9], our near-horizon solutions satisfy its eqs. (4.11) and (4.12) to 9.17×10^{-14} and 6.66×10^{-16} , respectively. This comparison checks the coupling conversion $\alpha_{\text{ref.}} = 4\alpha$.¹

Internal thermodynamic checks. At 10 selected states we evaluate the first law in the spin direction at fixed coupling:

$$\partial_q M - T \partial_q S - 2\Omega_H \partial_q J = 0.$$

Its worst residual is 1.35×10^{-8} relative to the largest term. We also evaluate $-T \partial_\alpha S|_{M,J}$ using coupling and spin responses and the scale direction. It agrees with Ψ to 4.84×10^{-8} relative. This is an internal check constructed from the same observables, not an independent test of the

¹A separate reconstruction of the non-extremal family gave an outer ergosurface radius 1.102101, compared with 1.104 quoted in [9], at $\alpha = 1/2$ in our convention. The discrepancy remains unresolved and is not used as evidence for an erratum. The potential calculation does not use the ergosurface radius.

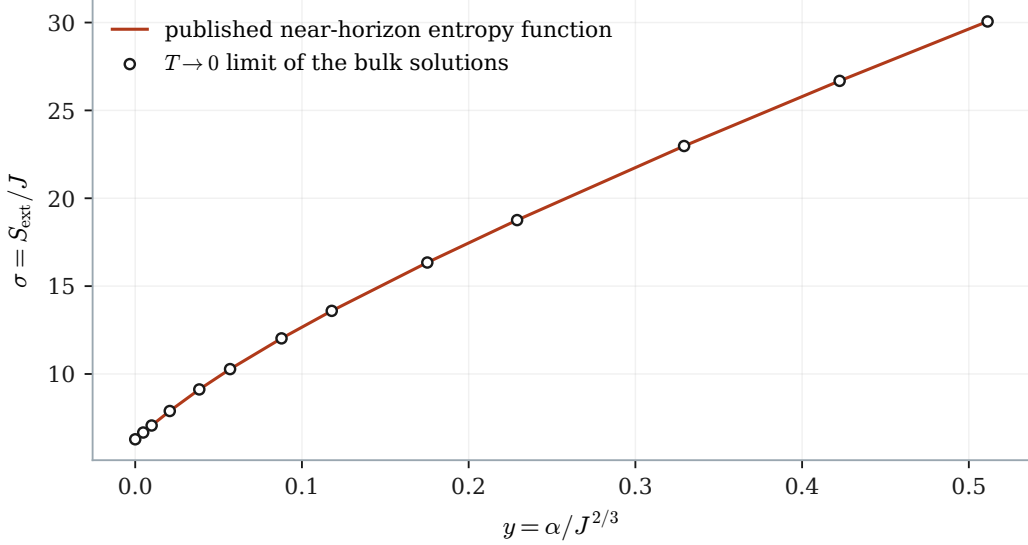


Figure 3. Extremal entropy per angular momentum. The line joins values from the near-horizon entropy function [9]; the points are extrapolations of bulk solutions to zero temperature.

first law. No degeneracy is implied by the static value $J = 0$: generically J vanishes linearly with spin and its spin derivative there is nonzero.

The Jacobian condition number at these selected states reaches 5.4×10^7 . Its role and the limits of this diagnostic are discussed in appendix A.

6 Discussion

The observable calculated here is the response of mass to the Gauss–Bonnet coupling at fixed entropy and angular momenta. Its zero-temperature limit gives the response of the extremal mass at fixed angular momenta, provided the branch has a regular thermodynamic limit. The numerical evidence supports a positive, non-monotonic slope over the sampled finite-coupling range. The positive initial slope was already known; the substantial decrease and subsequent increase are the finite-coupling behaviour investigated here.

The distinction between a slope and an accumulated mass change matters. At fixed J , the mass keeps increasing in the range measured. Its rate of increase reaches a lowest sampled value near $\alpha = 0.15$ in units $r_H = 1$, corresponding to $x = 0.1669$, and then recovers. By $x = 0.0871$ the slope is 1.3260, below half the perturbative value. The data do not establish its sign or shape outside this interval.

Positive Ψ also implies that entropy decreases with α at fixed (M, J) for positive-temperature states in that region. Meanwhile the entropy per spin along the extremal branch grows by a factor 4.8 in the sampled range. There is no contradiction: extremal states at fixed J have different masses as α changes.

There is another constraint distinction at low spin. Holding the coordinate radius r_H fixed does not hold the entropy fixed. At $\alpha = 1/2$ the measured mass initially decreases slightly with spin before increasing. The first law permits this when the negative entropy contribution exceeds the rotational contribution. The parity of $S(q)$ and $J(q)$ does not determine the sign of $\partial_q M$. We therefore do not interpret a mass below the static value at the same r_H as an error or use that static value as a lower bound on the rotating family. The published bounds $x < 1$ and $j \leq 1$ [11] are different statements: our largest scaled spin is 1.000000, and the extremal value decreases to 0.66328 at the largest sampled coupling.

The calculation treats EGB as a classical theory with its specified action. Second-order equations do not by themselves establish existence, uniqueness, stability or a well-posed problem at every coupling. We make claims only for the constructed branch and its extrapolated endpoints. In a gravitational effective field theory, terms beyond Gauss–Bonnet generically contribute at the same order as the higher powers of α retained by this finite-coupling calculation. Its non-monotonicity is consequently not a universal prediction of a derivative expansion. Causality restrictions on higher-curvature graviton interactions provide an additional reason to distinguish the classical truncation from an ultraviolet completion [19].

The main limitations are the extremal extrapolation, the incomplete coupling coverage and the equal-spin restriction. The displayed bands measure sensitivity to specified fitting choices; they are not rigorous numerical error bounds. The mass-curve consistency check shares systematic errors with the potential calculation, while the near-horizon check determines entropy rather than mass. The near-horizon geometry discards the asymptotic normalisation needed to recover the global mass by itself. A direct construction of extremal solutions with response measurements, additional resolution studies, and a comparison with an independent solver would strengthen the quantitative determination. A second-order perturbative calculation would also provide a useful test of the initial departure from $\Psi_{\text{ext}} = \pi$.

A Numerical construction and sensitivity analysis

Radial discretisation. We set $r_H = 1$, introduce $z = 1/r$ and collocate in $\chi = 1 - z$. The horizon is at $\chi = 0$ and infinity at $\chi = 1$. The unknown amplitudes factor out the boundary behaviour:

$$b = (1 - z^2)B, \quad f = (1 - z^2)F, \quad h = \frac{1 + z^4 H}{z^2}, \quad w = z^4 W.$$

They are represented at Chebyshev–Lobatto nodes. At infinity we impose $B = F = 1$, $H' = W' = 0$. At the horizon, where the solved interior equations are singular, we impose three regularity relations and $W(0) = \Omega_H$. The equations and horizon relations are generated symbolically. Damped Newton iteration solves the collocation system. Production walks use $N = 64$, with $N = 96$ available when acceptance requires refinement; $N = 48$ is used for selected comparison walks.

Jacobian and response. The production Newton and response solves use the same chain-rule assembly of $\partial R_N / \partial u$. Local partial derivatives are computed by complex step: seven vectorised partials of the interior equations and nine of the horizon relations. The coupling and angular-velocity derivatives are computed by complex step in the respective parameter, with $\epsilon = 10^{-20}$. The rational residual is evaluated using complex arithmetic without non-analytic operations on the perturbed variables. The assembled Jacobian is checked entrywise against a dense complex-step perturbation of every nodal unknown, including profiles that do not solve the equations. This checks the chain-rule assembly; both implementations still use complex-step differentiation.

Continuation. Starting from Myers–Perry, we continue in coupling and spin. The tangent predictor $u + \Delta\alpha \partial_\alpha u + \Delta q \partial_q u$ provides the initial guess for each nonlinear solve. In the tested near-extremal sequences it reaches states at which reusing the previous solution failed to converge. The predictor is only an initial guess: every retained state must satisfy the same acceptance checks.

Acceptance and conditioning. An evaluator separate from the generated radial equations constructs curvature tensors from the metric and checks $G_{\mu\nu} + \alpha H_{\mu\nu}$ at 51 off-grid points. We

divide the maximum absolute residual over these points by the maximum reference scale over the same points. The reference scale is the largest of the Einstein and Gauss–Bonnet tensor contributions and the square root of the absolute Kretschmann invariant. The relative threshold is 10^{-8} ; the absolute residual is recorded as well. The largest relative residual over the accepted states is 9.76×10^{-9} , and the largest relative Smarr residual is 1.64×10^{-7} . These residuals diagnose equation closure; they are not bounds on the error of an extracted observable.

The largest condition number in the selected 10 checks is 5.4×10^7 . Multiplying a condition number by machine precision estimates possible error amplification, not the number of correct digits in Ψ . These selected checks do not bound conditioning at all near-extremal states or at every resolution.

Asymptotic charge extraction. The coefficients in (6) are obtained by fitting compactified amplitudes in z^2 . The three windows and degrees are $z \in [0.03, 0.12]$ at degree 4, $z \in [0.04, 0.16]$ at degree 4, $z \in [0.03, 0.12]$ at degree 3. The first is the production choice; the other two measure sensitivity to that choice. We also compare masses extracted from the b and f asymptotics. The largest recorded relative diagnostic is 1.31×10^{-8} . All these extractions can share a bias, so a small spread does not certify their absolute accuracy. At $\alpha = 0$, comparison with the analytic $M = 3\pi r_H^2/[8(1 - q^2)]$ gives a worst relative deviation 3.27×10^{-13} through $q = 0.7$. The static finite-coupling comparison is in section 4.1. We have no corresponding analytic rotating mass at finite coupling.

Central extrapolation. At each coupling, states are sorted by the physical temperature T in units $r_H = 1$. The central estimates of M, J, S, Ψ are the intercepts of cubic fits through the six coldest states. The original degree checks use linear, quadratic and cubic fits through four, five and six states, respectively. The invariants y, μ, σ are then formed from the extrapolated charges. The use of unscaled T here is distinct from the use of $\tau = TJ^{1/3}$ to report the distance to extremality.

Fit-sensitivity envelopes. For the potential we recompute fits on the six coldest states and on up to twelve coldest states, using both T and τ . For each coordinate and window we use polynomials of degree one, two and three and the forms

$$c_0 + c_1 v + c_1/2 \sqrt{v}, \quad c_0 + c_1 v + c_2 v^2 + c_{\log} v^2 \log v,$$

where v is the chosen coordinate divided by its largest value in the fitting window. This rescaling improves conditioning. The half-width displayed in table 2 and figure 2 is the largest deviation of any of these intercepts from the production central estimate, or the original degree spread if larger. The central values are unchanged by this reanalysis.

Table 3 shows the largest changes from the square-root and logarithmic models across the two coordinates and windows. These terms are sensitivity probes, not a derivation of the near-extremal expansion. Small changes for a given probe do not exclude other non-analytic behaviour. The envelope also includes the polynomial window and coordinate changes; it has no assigned statistical confidence level and does not bound biases shared by all fits.

Repeating three walks at the comparison resolution changes the extrapolated potential by at most 2.56×10^{-5} and μ by at most 6.03×10^{-6} . This supports resolution stability at those couplings, but does not replace a systematic convergence study of every near-extremal endpoint. The integral comparison in (17) has its own interpolation error and correlations with both sets of estimates; we report its discrepancy separately instead of interpreting it as coverage by independent error bars.

α	$T_{\min}$	$T_{\max}/T_{\min}$	square-root shift	logarithmic shift
0	1.08×10^{-5}	3919	1.07×10^{-2}	1.11×10^{-4}
0.005	2.40×10^{-3}	24	1.12×10^{-1}	2.88×10^{-3}
0.01	1.21×10^{-3}	49	8.81×10^{-2}	2.37×10^{-3}
0.02	4.73×10^{-3}	13	1.60×10^{-1}	8.97×10^{-3}
0.035	3.96×10^{-3}	15	1.32×10^{-1}	7.61×10^{-3}
0.05	4.80×10^{-4}	86	1.86×10^{-2}	3.59×10^{-4}
0.075	4.75×10^{-4}	86	2.20×10^{-2}	4.31×10^{-4}
0.1	4.17×10^{-4}	94	2.10×10^{-2}	4.17×10^{-4}
0.15	2.78×10^{-4}	107	1.30×10^{-2}	2.01×10^{-4}
0.2	1.55×10^{-4}	183	1.12×10^{-2}	1.85×10^{-4}
0.3	1.11×10^{-4}	173	6.16×10^{-3}	8.20×10^{-5}
0.4	5.20×10^{-5}	392	6.96×10^{-3}	1.29×10^{-4}
0.5	3.38×10^{-6}	2900	1.38×10^{-3}	1.24×10^{-5}

Table 3. Sensitivity of the potential intercept to non-polynomial fit models. The temperature range is that of the up-to-twelve-point window. Each shift is the largest absolute difference from the production central value over both windows and both coordinates.

Data and code. The supplementary table contains the 356 accepted states, their observables and acceptance diagnostics. The accompanying source tree contains the solver, symbolic derivation scripts, tests and figure-generation script. The latter also exports the fitted intercepts and sensitivity envelopes used in this version, together with source hashes. These materials document the calculation; this manuscript does not claim an existing public archival DOI.

B The near-horizon entropy function

The comparison in section 5 uses the extremal entropy obtained from the near-horizon geometry, following section 4.2 of [9]. We summarise our rederivation, which is independent of that reference in execution though not in content.

For equal angular momenta the extremal near-horizon geometry is homogeneous, with isometry group $SO(2, 1) \times SU(2) \times U(1)$; see [18] for the general classification of such geometries. It can be written in terms of the $SU(2)$ invariant one-forms σ_i as

$$ds^2 = v_1 \left(-r^2 dt^2 + \frac{dr^2}{r^2} \right) + \frac{v_2}{4} (\sigma_1^2 + \sigma_2^2) + \frac{v_3}{4} (\sigma_3 + k r dt)^2, \quad (18)$$

with $v_{1,2,3}$ and k constants. On (18) both R and $\mathcal{L}_{\text{GB}}$ are constant, and Sen’s entropy function [10] reduces the problem to algebra: with

$$f(v, k) = \int d\theta d\varphi d\psi \sqrt{-g} \frac{R + \alpha \mathcal{L}_{\text{GB}}}{16\pi} = \frac{\pi}{8} v_1 v_2 \sqrt{v_3} (R + \alpha \mathcal{L}_{\text{GB}}), \quad (19)$$

the equations of motion are $\partial f / \partial v_i = 0$, the angular momentum is $J = \partial f / \partial k$, and the entropy is the Legendre transform $S_{\text{ext}} = 2\pi(kJ - f)$ evaluated at the extremum.

We derived R and $\mathcal{L}_{\text{GB}}$ on (18) symbolically and checked them against the same numerical curvature evaluator used for acceptance; we then verified that the extremal solution satisfies the full field equations $G_{\mu\nu} + \alpha H_{\mu\nu} = 0$ component by component, since extremising a reduced functional and solving the equations are not the same statement. At $\alpha = 0$ the system solves exactly, $k = 1$, $v_2 = 4v_1$, $v_3 = 8v_1$, giving $S_{\text{ext}} = 2\pi J$, which is extremal Myers–Perry ($S = 2\pi^2 a^3$, $J = \pi a^3$ at $r_+ = a$) with nothing tuned: the $1/16\pi G$ of the action and the 2π of the Legendre transform were both fixed beforehand. At finite coupling the branch is a root of an explicit

polynomial system, and it reproduces eqs. (4.11) and (4.12) of [9] to the precision quoted in section 5.

Under $v_i \rightarrow \lambda v_i$, $\alpha \rightarrow \lambda \alpha$ both J and S_{ext} scale as $\lambda^{3/2}$, so $\sigma = S_{\text{ext}}/J$ and $y = \alpha/J^{2/3}$ are invariant and are what figure 3 compares. We emphasise that this appendix produces no mass: the near-horizon geometry of an extremal black hole determines its entropy as a function of its charges and no more, a point made in [9] itself, where the absence of the corresponding $E(J)$ diagram is noted as an open problem.

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